2026 Tanner Crab Transition To GMACS

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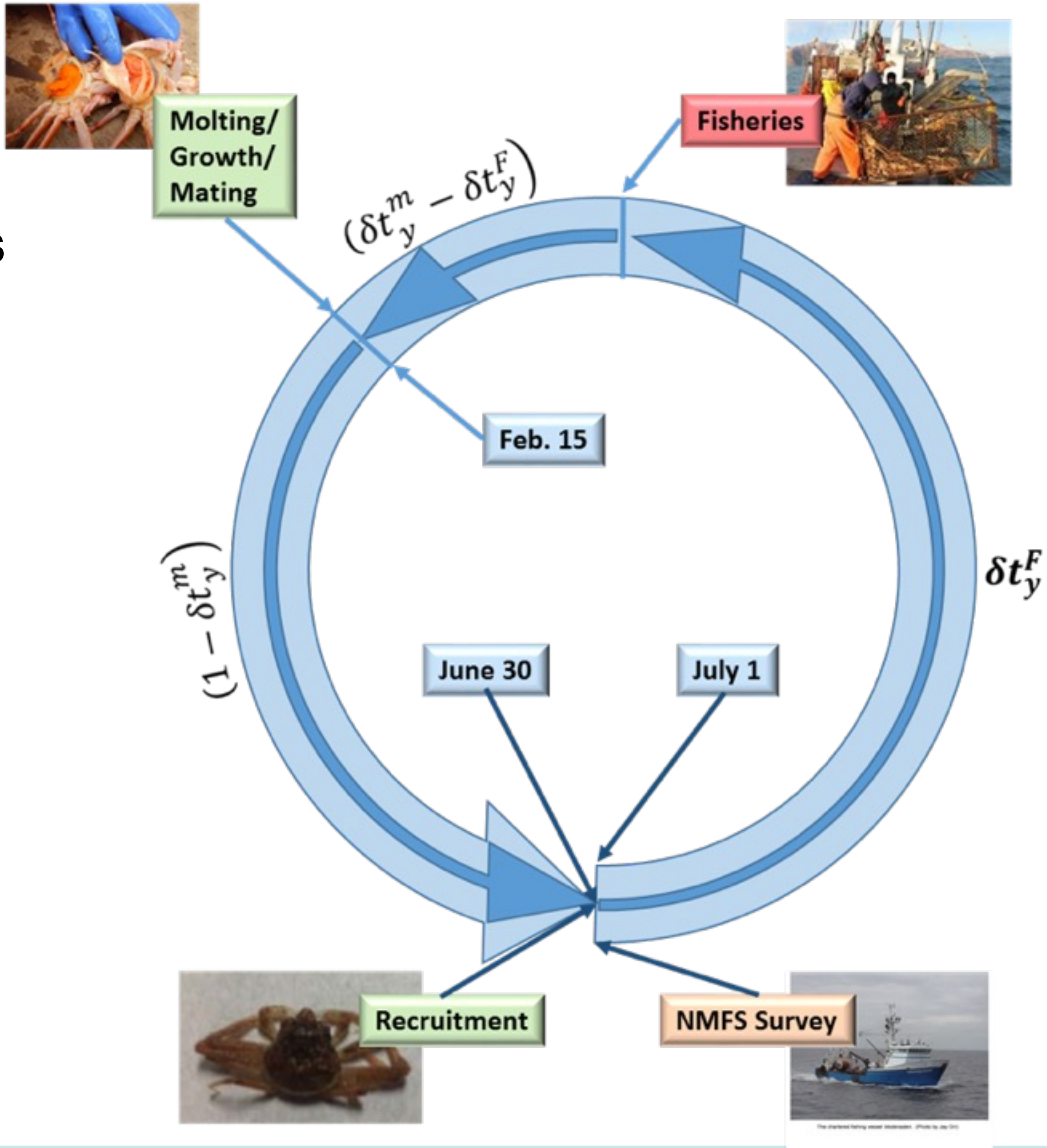
AFSC/NMFS/NOAA

Tier 3 Assessment Model

- Tier 3 size-structured model
  - Survey data
    - NMFS EBS shelf survey: 1975-present
    - BSFRF 2013-2018 side-by-side haul studies
  - Fishery data
    - directed fishery (areas combined)
      - retained catch
      - total catch
    - bycatch in
      - snow crab fishery
      - BBRKC fishery
      - groundfish fisheries
  - Estimates:
    - Annual recruitment
    - Annual numbers-at-size (M,F)
    - mature biomass (MMB, MFB)
  - Determines:
    - F_{MSY}, B_{MSY}, F_{OFL}, OFL



Population dynamics (1)

Population abundance at start of year $n_{y,x,m,s,z}$

- y: year
- x: sex
- m: maturity state (immature, mature)
- s: shell condition (new shell, old shell)
- z: size (5 mm CW bins, cutpoints: 25-185)

Step A1.1: Survival prior to fisheries $n^1_{y,x,m,s,z} = e^{-M_{y,x,m,s,z} \cdot \delta t^F_y} \cdot n_{y,x,m,s,z}$

Step A1.2: Prosecution of the fisheries $n^2_{y,x,m,s,z} = e^{-F^T_{y,x,m,s,z}} \cdot n^1_{y,x,m,s,z}$

Step A1.3: Survival after fisheries to time of molting/growth/mating $n^3_{y,x,m,s,z} = e^{-M_{y,x,m,s,z} \cdot \left(\delta t^m_y - \delta t^F_y\right)} \cdot n^2_{y,x,m,s,z}$

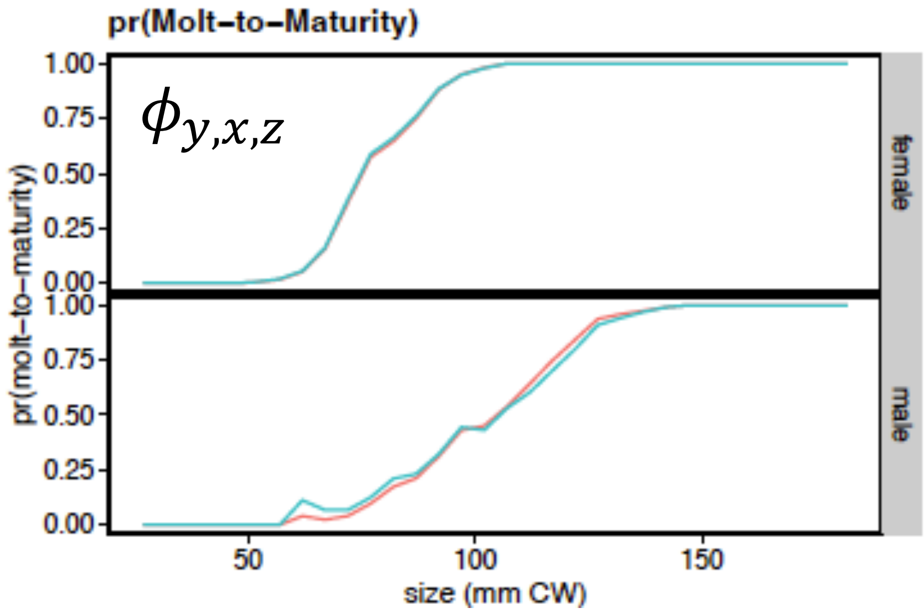
Population dynamics (2)

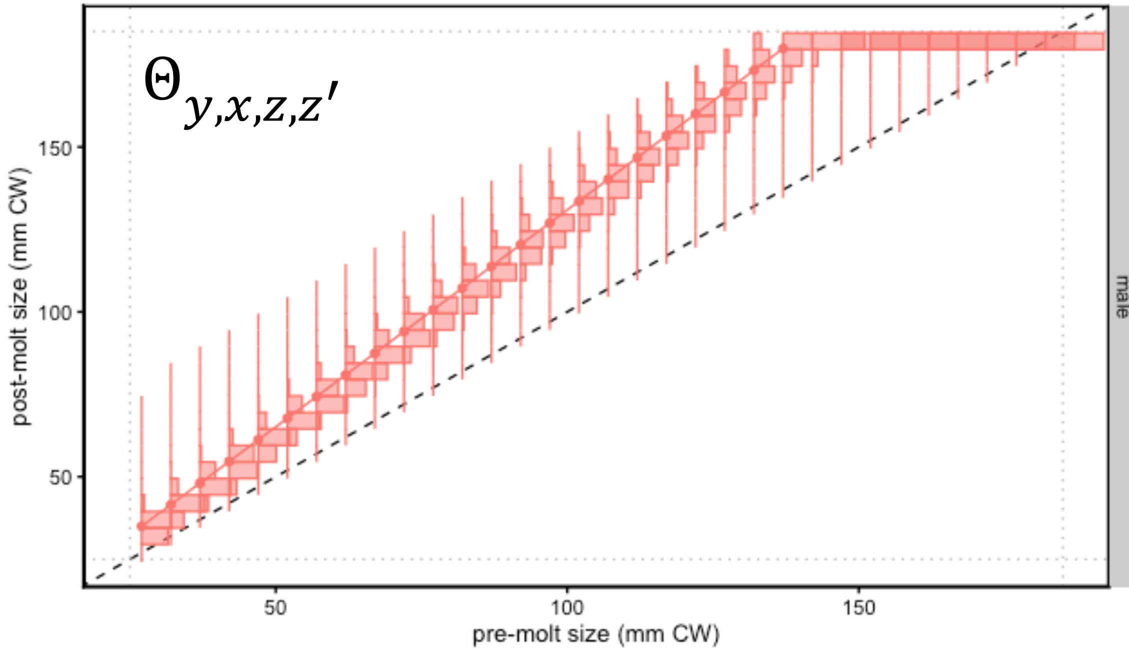
Step A1.4: Molting, growth, and maturation

$$n^4_{y,x,IMM,NS,z} = (1-\phi_{y,x,z}) \cdot \sum_{z'} \Theta_{y,x,z,z'} \cdot n^3_{y,x,IMM,NS,z'}$$

$$n^4_{y,x,MAT,NS,z} = \phi_{y,x,z} \cdot \sum_{z'} \Theta_{y,x,z,z'} \cdot n^3_{y,x,IMM,NS,z'}$$

$$n^4_{y,x,MAT,OS,z} = n^3_{y,x,MAT,OS,z} + n^3_{y,x,MAT,NS,z}$$

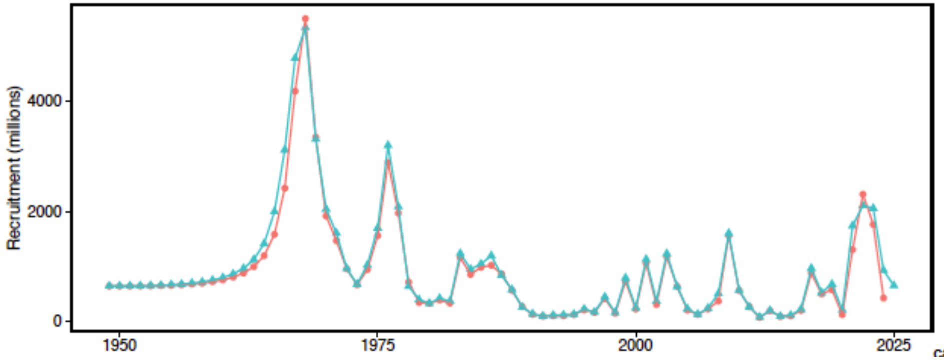




Population dynamics (3)

Step A1.5: Survival to end of year, recruitment, and update to start of next year

$$n_{y+1,x,m,s,z} = \begin{cases} e^{-M_{y,x,IMM,NS,z}\cdot(1-\delta t_y^m)} \cdot n^4_{y,x,IMM,NS,z} + R_{y,x,z} & m = IMM, s = NS \\ e^{-M_{y,x,m,s,z}\cdot(1-\delta t_y^m)} \cdot n^4_{y,x,m,s,z} & otherwise \end{cases}$$



$$R_{y,x,z} = \dot{R}_y \cdot \ddot{R}_{y,x} \cdot \dddot{R}_{y,z}$$

$$\dot{R}_y = e^{pLnR_t+\delta R_{t,y}} \quad y \in t$$

$$\ddot{R}_{y,x} = 0.5$$

$$\dddot{R}_{t,z} = c^{-1} \cdot \Delta_z^{\frac{\alpha_t}{\beta_t}-1} \cdot e^{-\frac{\Delta_z}{\beta_t}}$$

Fishery dynamics (1)

"Instantaneous" fisheries (no M)

Fishing mortality: $F_{f,y,x,m,s,z} = \left[h_{f,t} \cdot \left(1 - \rho_{f,y,x,m,s,z}\right) + \rho_{f,y,x,m,s,z}\right] \cdot \phi_{f,y,x,m,s,z}$

Handling (discard) mortality: $h_{f,t}$

Retention probability: $\rho_{f,y,x,m,s,z}$

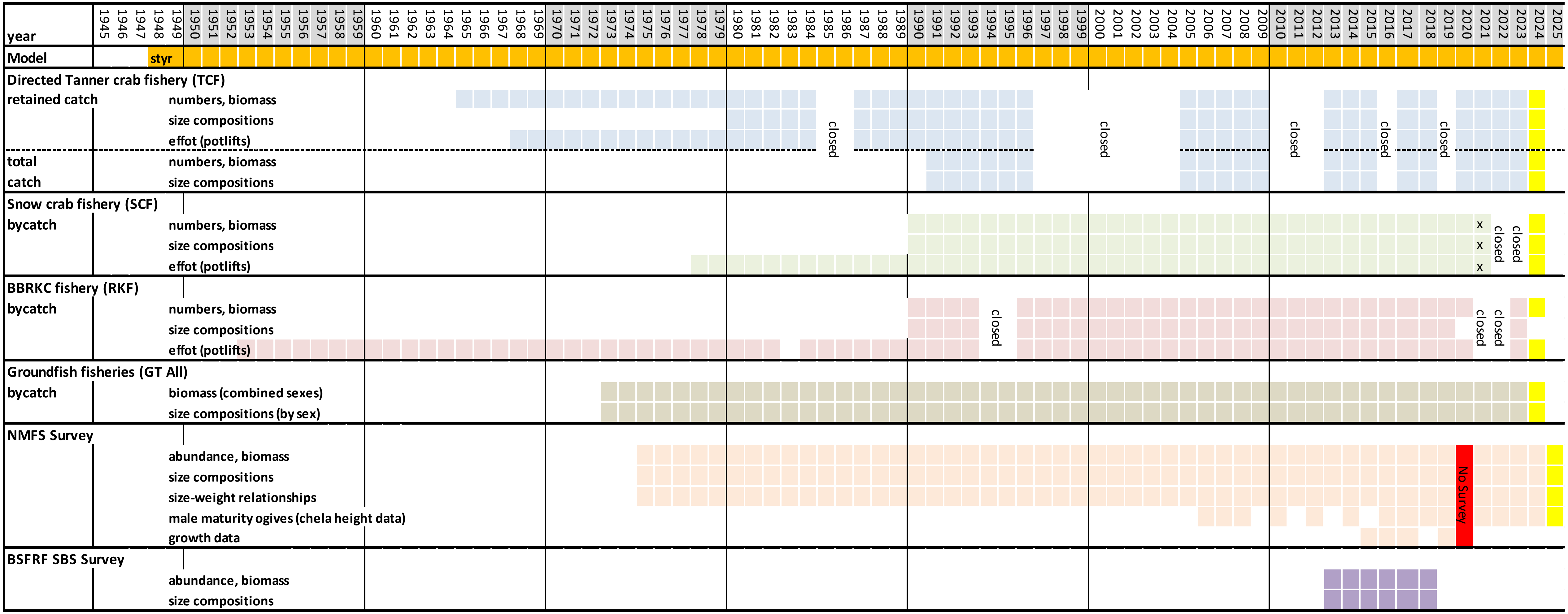
Capture rate: $\phi_{f,y,x,m,s,z} = \phi_{f,y,x,m,s} \cdot S_{f,y,x,m,s,z}$

Numbers caught: $c_{f,y,x,m,s,z} = \frac{\phi_{f,y,x,m,s,z}}{F^T_{y,x,m,s,z}} \cdot \left[1 - e^{-F^T_{y,x,m,s,z}}\right] \cdot n_{y,x,m,s,z}$

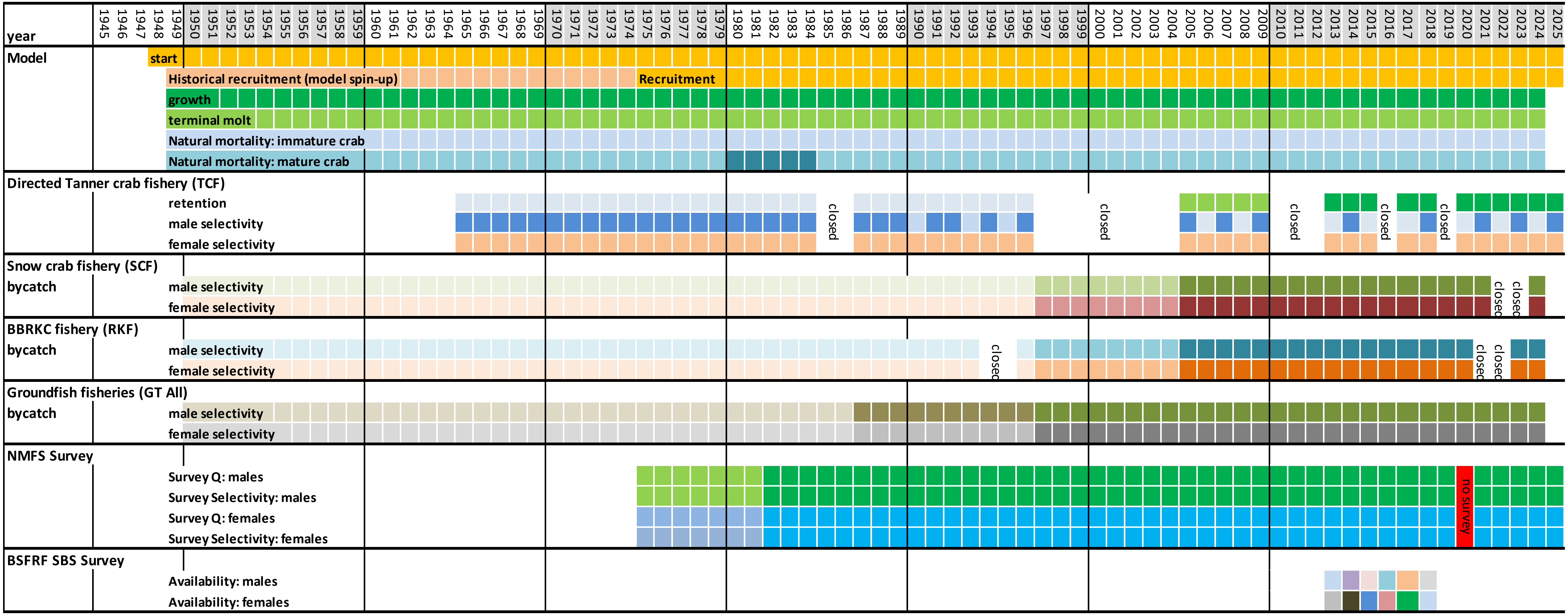
Numbers retained: $r_{f,y,x,m,s,z} = \frac{\rho_{f,y,x,m,s,z} \cdot \phi_{f,y,x,m,s,z}}{F^T_{y,x,m,s,z}} \cdot \left[1 - e^{-F^T_{y,x,m,s,z}}\right] \cdot n_{y,x,m,s,z}$

discard mortality: $dm_{f,y,x,m,s,z} = \frac{h_{f,y} \cdot \left(1 - \rho_{f,y,x,m,s,z}\right) \cdot \phi_{f,y,x,m,s,z}}{F^T_{y,x,m,s,z}} \cdot \left[1 - e^{-F^T_{y,x,m,s,z}}\right] \cdot n_{y,x,m,s,z}$

Assessment time frames: data



Assessment time frames: model processes



Population dynamics

process	time blocks	time blocks	22.03d5 description
Population rates and quantities			
Population built from annual recruitment			
Recruitment	1949+	1949-1974	ln-scale mean + annual devs constrained as AR1 process
		1975+	ln-scale mean + annual devs
	1949+	1949+	sigma-R fixed, sex ratio fixed at 1:1
Growth	1949+	1949+	sex-specific
			mean post-molt size: power function of pre-molt size
			post-molt size: gamma distribution conditioned on pre-molt size
Maturity	1949+	1949+	sex-specific
			size-specific probability of terminal molt
			logit-scale parameterization
Natural mortalty	1949-1979,	1949-1979,	estimated sex/maturity state-specific multipliers on base rate
	1985+	1985+	priors on multipliers based on uncertainty in max age
	1980-1984	1980-1984	estimated "enhanced mortality" period multipliers

Fisheries

Fishery/process	time blocks	22.03d5 description
RKF	bycatch in BBRKC fishery	
capture rates	pre-1952	nominal rate on males
	1953-1991	extrapolated from effort
	1992+	male ln-scale mean + annual devs(*)
	1949+	ln-scale female offset
male selectivity	1949-1996	ascending normal, asymptote fixed
	1997-2004	ascending normal, asymptote fixed
	2005+	ascending normal, asymptote fixed
female selectivity	1949-1996	ascending normal, asymptote fixed
	1997-2004	ascending normal
	2005+	ascending normal
GF All	bycatch in groundfish fisheries	
capture rates	pre-1973	male ln-scale mean from 1973+
	1973+	male ln-scale mean + annual devs
	1973+	ln-scale female offset
male selectivity	1949-1986	ascending logistic
	1987-1996	ascending logistic
	1997+	ascending logistic
female selectivity	1949-1986	ascending logistic
	1987-1996	ascending logistic
	1997+	ascending logistic

Fishery/process	time blocks	22.03d5 description
TCF	directed Tanner crab fishery	
capture rates	pre-1965	male nominal rate
	1965+	male ln-scale mean + annual devs
	1949+	ln-scale female offset
male selectivity	1949-1990	ascending logistic
	1991-1996	annually-varying ascending logistic
	2005+	annually-varying ascending logistic
female selectivity	1949+	ascending logistic
male retention	1949-1990; 1991-1996; 2005-2009; 2013+	ascending logistic
% retained	pre-1988	fixed at 100%
	1991-1996	fixed at 100%
	2005-2009	fixed at 100%
	2013+	fixed at 100%
SCF	bycatch in snow crab fishery	
capture rates	pre-1978	nominal rate on males
	1979-1991	extrapolated from effort
	1992+	male ln-scale mean + annual devs
	1949+	ln-scale female offset
male selectivity	1949-1996	dome-shaped (double normal) --plateau width fixed to 0 --descending limb width fixed to 1
	1997-2004	dome-shaped (double normal)
	2005+	dome-shaped (double normal)
female selectivity	1949-1996	ascending logistic
	1997-2004	ascending logistic
	2005+	ascending logistic

Surveys

Survey/process	time blocks	22.03d5 description
NMFS EBS trawl survey		
male survey q	1975-1981	ln-scale
	1982+	ln-scale w/ prior based on Somerton's underbag experiment
female survey q	1975-1981	ln-scale
	1982+	ln-scale w/ prior based on Somerton's underbag experiment
male selectivity	1975-1981	ascending normal, fixed fully-selected size at 180
	1982+	ascending normal, fixed fully-selected size at 180
female selectivity	1975-1981	ascending normal, fixed fully-selected size at 130
	1982+	ascending normal, fixed fully-selected size at 130
BSFRF SBS trawl surveys		
male catchability	2013-2018	fixed at 1 for all sizes
male availability	2013-2018	empirically-determined outside the model
female catchability	2013-2018	fixed at 1 for all sizes
female availability	2013-2018	empirically-determined outside the model

Likelihoods

		22.03d5		
Component	Type	included in optimization	Fits	Likelihood distribution
TCF: retained catch	biomass	yes	males only	lognormal
	size comp.s	yes	males only	multinomial
TCF: total catch	biomass	yes	total	lognormal
	size comp.s	yes	by sex (extended)	multinomial
SCF: total catch	biomass	yes	total	lognormal
	size comp.s	yes	by sex (extended)	multinomial
RKF: total catch	biomass	yes	total	lognormal
	size comp.s	yes	by sex (extended)	multinomial
GF All: total catch	abundance	yes	total	lognormal
	biomass	yes	total	lognormal
	size comp.s	yes	by sex	multinomial
NMFS "M" survey (males only, no maturity)	biomass	yes	males only	lognormal
	size comp.s	yes	males only	multinomial
NMFS "F" survey (females only, w/ maturity)	biomass	yes	by maturity classification	lognormal
	size comp.s	yes	by maturity classification	multinomial
BSFRF "M" survey (males only, no maturity)	biomass	yes	males only	lognormal
	size comp.s	yes	males only	D-M
BSFRF "F" survey (females only, w/ maturity)	biomass	yes	by maturity classification	lognormal
	size comp.s	yes	by maturity classification	D-M
growth data	EBS only	yes	by sex	gamma
male maturity ogive data	EBS only	yes	males only	binomial

Some Major Differences Between TCSAM02 and GMACS

TCSAM02	GMACS
terminal molt only	non- and terminal molt
fixed seasons	user-defined number of seasons
continuous fisheries in 0-length season	allows "instantaneous" fisheries
all fisheries in same season	fisheries can occur in different seasons
allows F's, S's, r's by x,m,s	allows F's, S's, r's by x only
fits maturity ogive data for Pr(Imm->Mat)	uses maturity ogive data as Pr(Imm->Mat)
can fit fishery catch data by x,m,s	can fit fishery data by x
can fit survey index data by x,m,s	can fit survey data by x,m
can fit size comps by x,m,s,z	can fit size comps by x,m,s,z
different parameterizations	
time blocks for average recruitment	no time blocks for average recruitment
different priors, penalties	

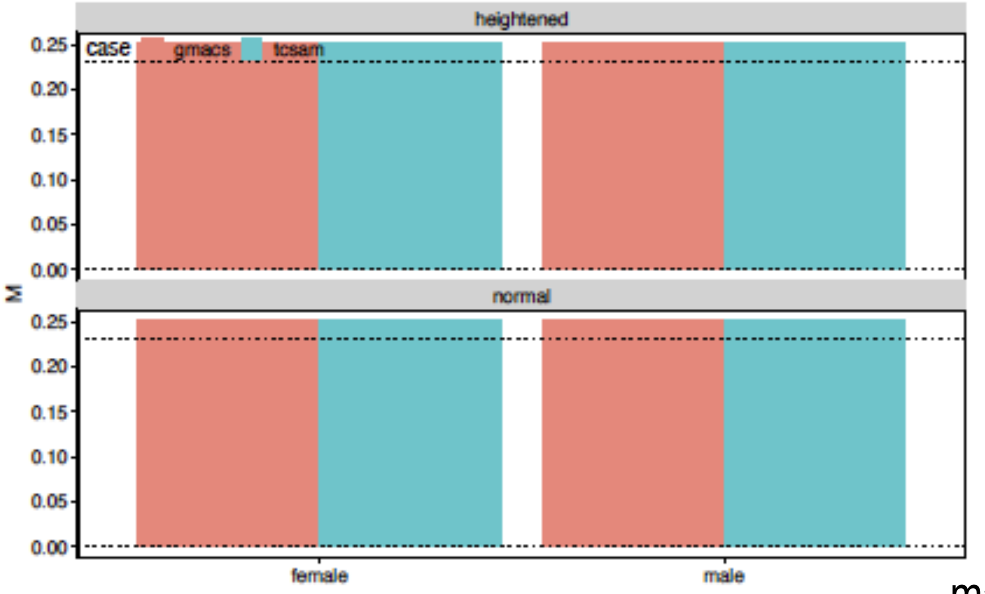
GMACS Modifications To Match TCSAM02

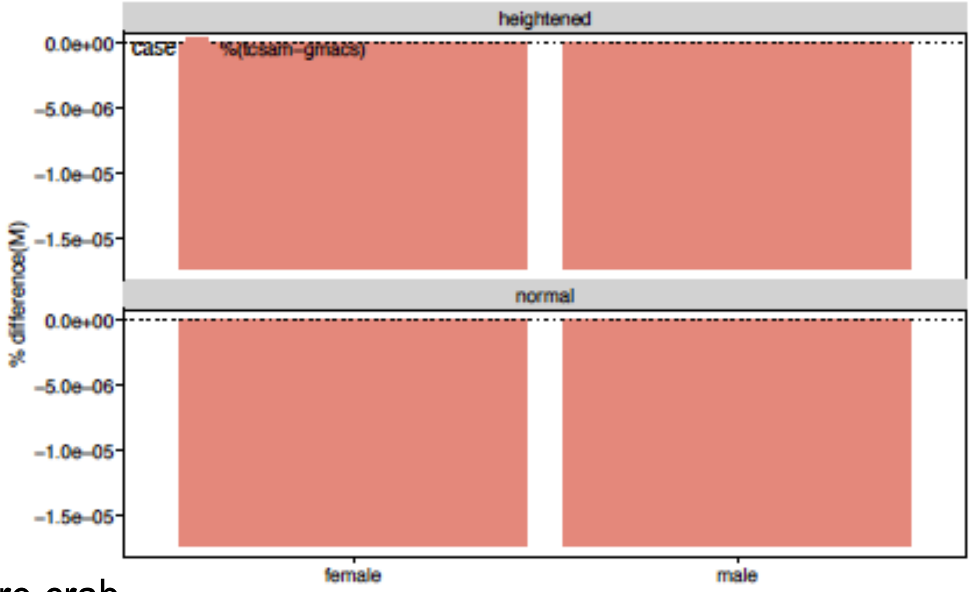
- *Added alternative multinomial likelihood as option to fit size comps (option 7) to match TCSAM02 calculation.*
- *Added male maturity data section to data file.*
- *Added "doTCSAM02EffExp" flag to data file (right after flag to read catch data in new format). If "on" (1), effort extrapolation for missing catch is based on average catch divided by average effort for years specified by new "avg_eff" column in new-format CatchDFs.*
- *Added growth transition options 9 and 10 to provide same functionality as TCSAM02*
- *Added alternative 4-parameter double normal selectivity option (SELEX_DOUBLENORM4a = 15) to match TCSAM02. The plateau is determined from a scale parameter and the distance between the max possible size and the size at the ascending limb.*
- *The growth options TCSAM02A and TCSAM02B (=9 and 10) that provide similar functionality as TCSAM02 for the growth transition matrices were reassigned and additional options TCSAM02C and TCSAM02D (11 and 12) were added. A and B apply to the case where the molt increment is assumed to be gamma-distributed while C and D apply to the case where postmolt size is gamma-distributed. All of the options limit growth to a maximum number of size bins but treat probabilities from the "excluded" size bins differently: A and C drop them, then normalize the remaining probabilities to sum to 1 whereas B and D treat the max size bin as an accumulator for "raw" probabilities at larger size bins before dropping them.*
- *Added new options for calculating/extending size comps to provide sample size scaling/no scaling options for input size comps.*
- *Fixed calculation of predicted fishery size comps in calc_predicted_composition by including fully-selected F's (ft(k,h,i,j)) in the calculations.*
- *Added new option for averaging fishing mortality rates before calculating the OFL*

Natural Mortality

immature crab

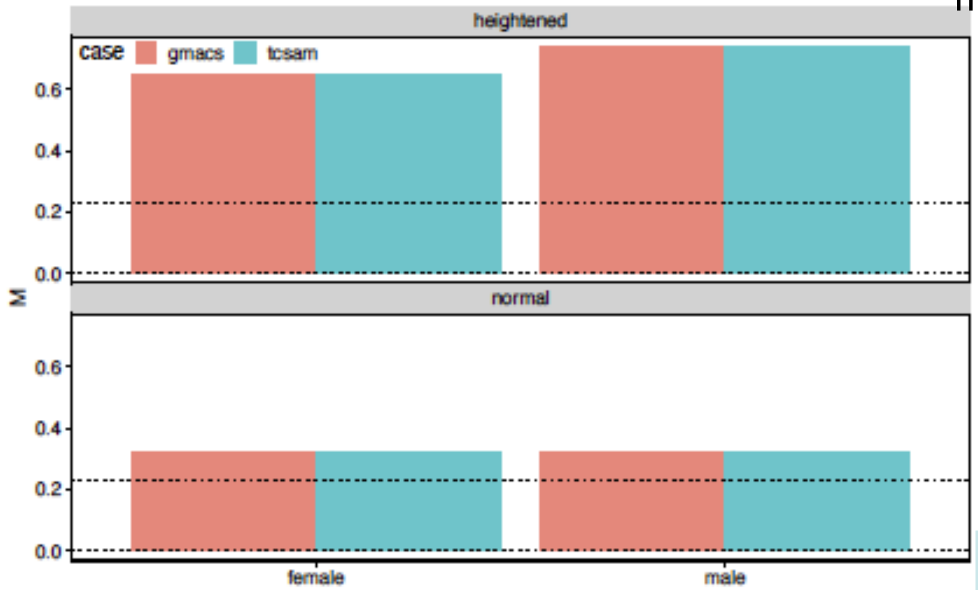
values

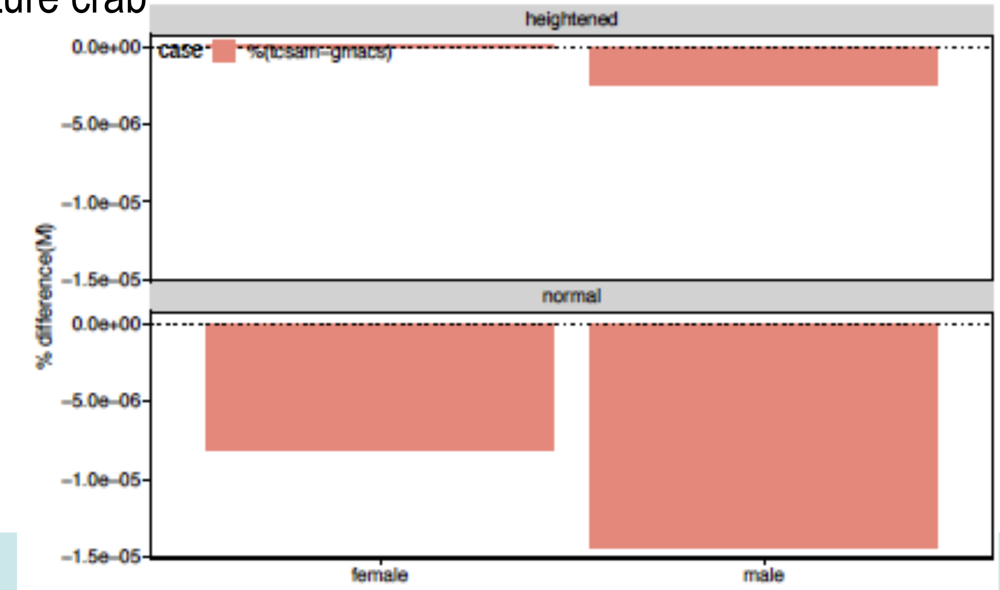




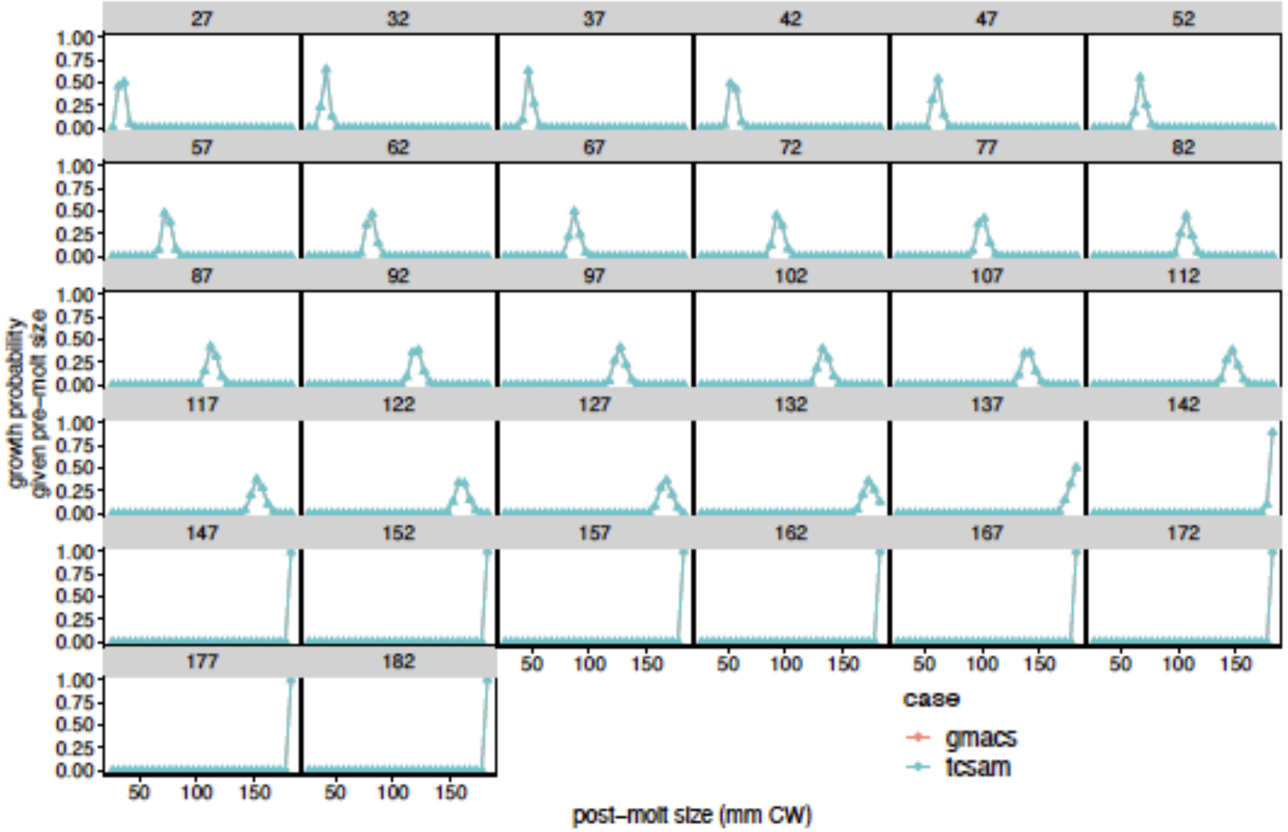
% differences

mature crab





Male growth



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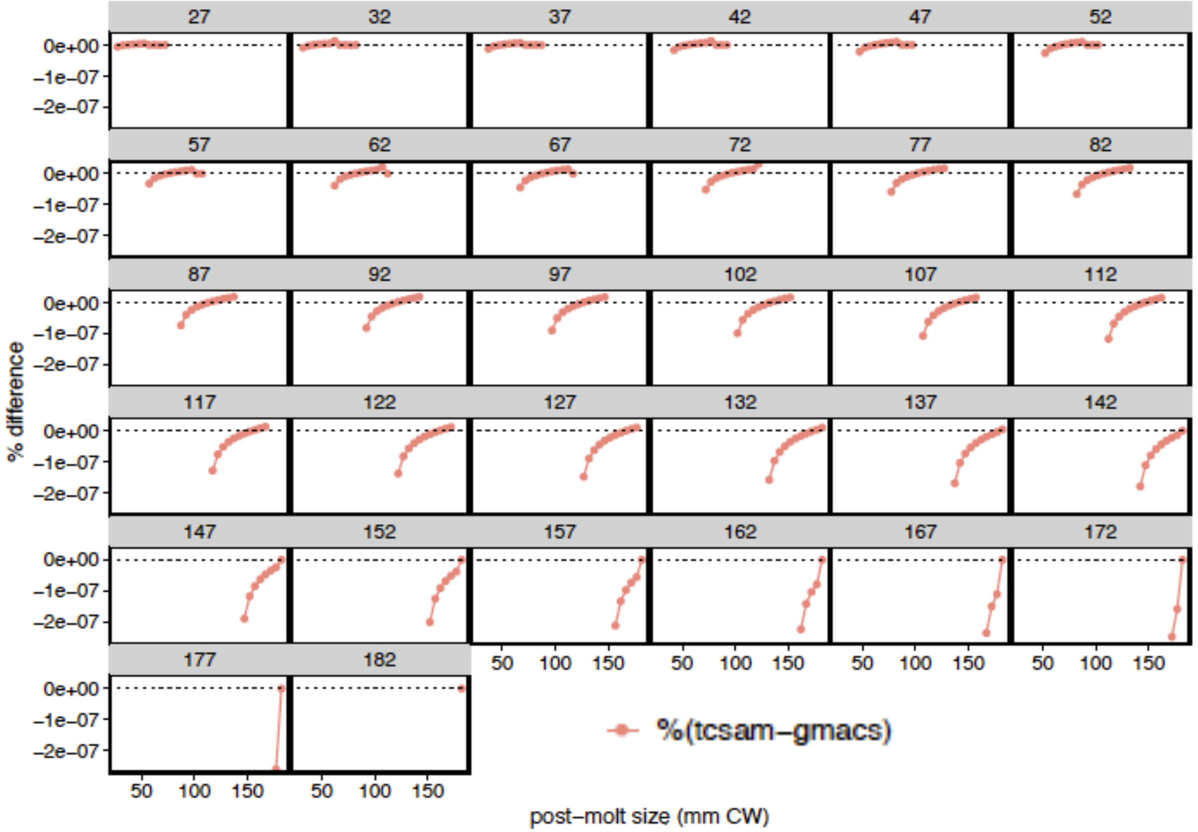
growth probability given pre-molt size
1.00
0.75
0.50
0.25
0.00
50
100
150
post-molt size (mm CW)
case
gmacs
tcsam



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177
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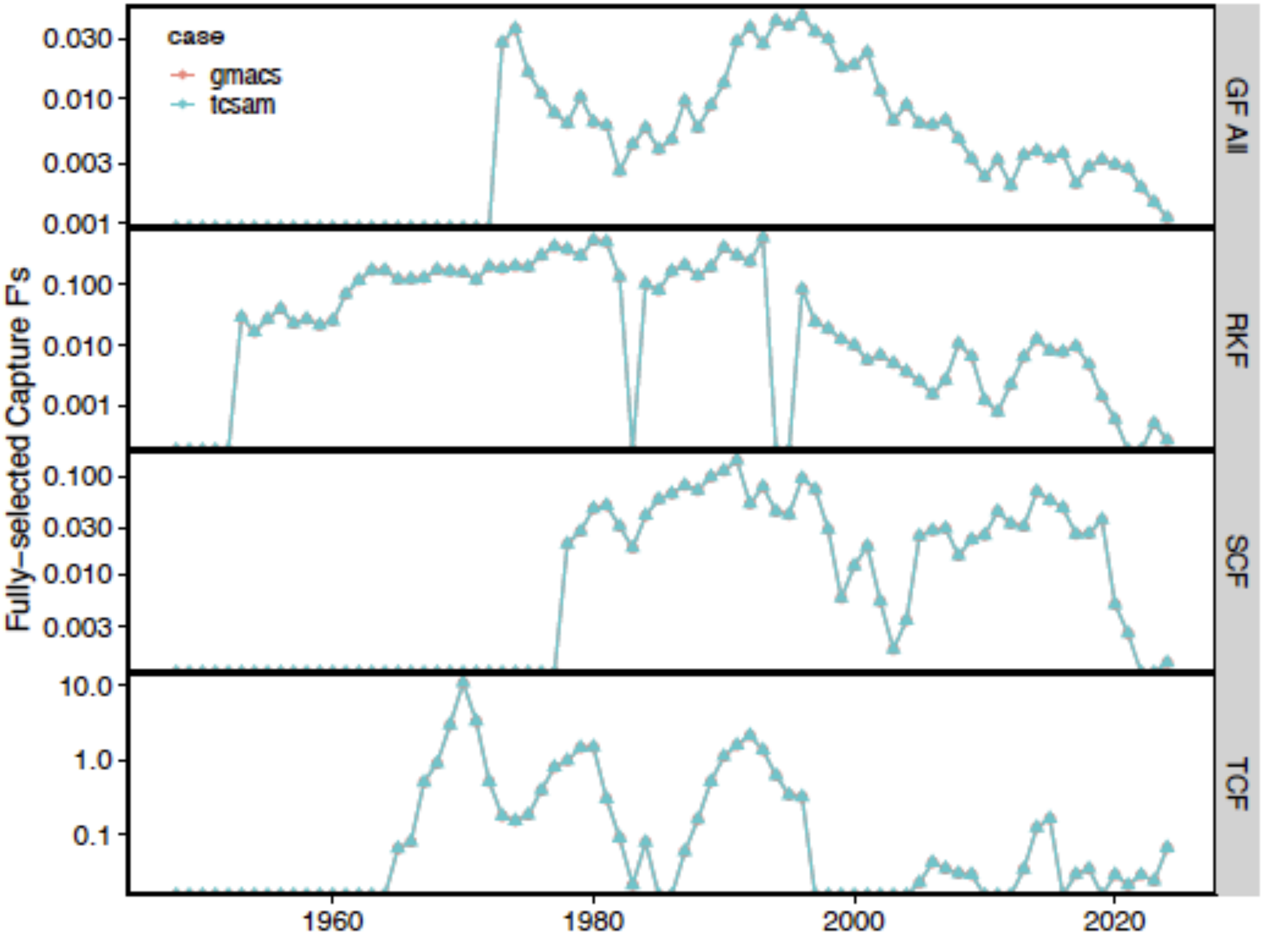
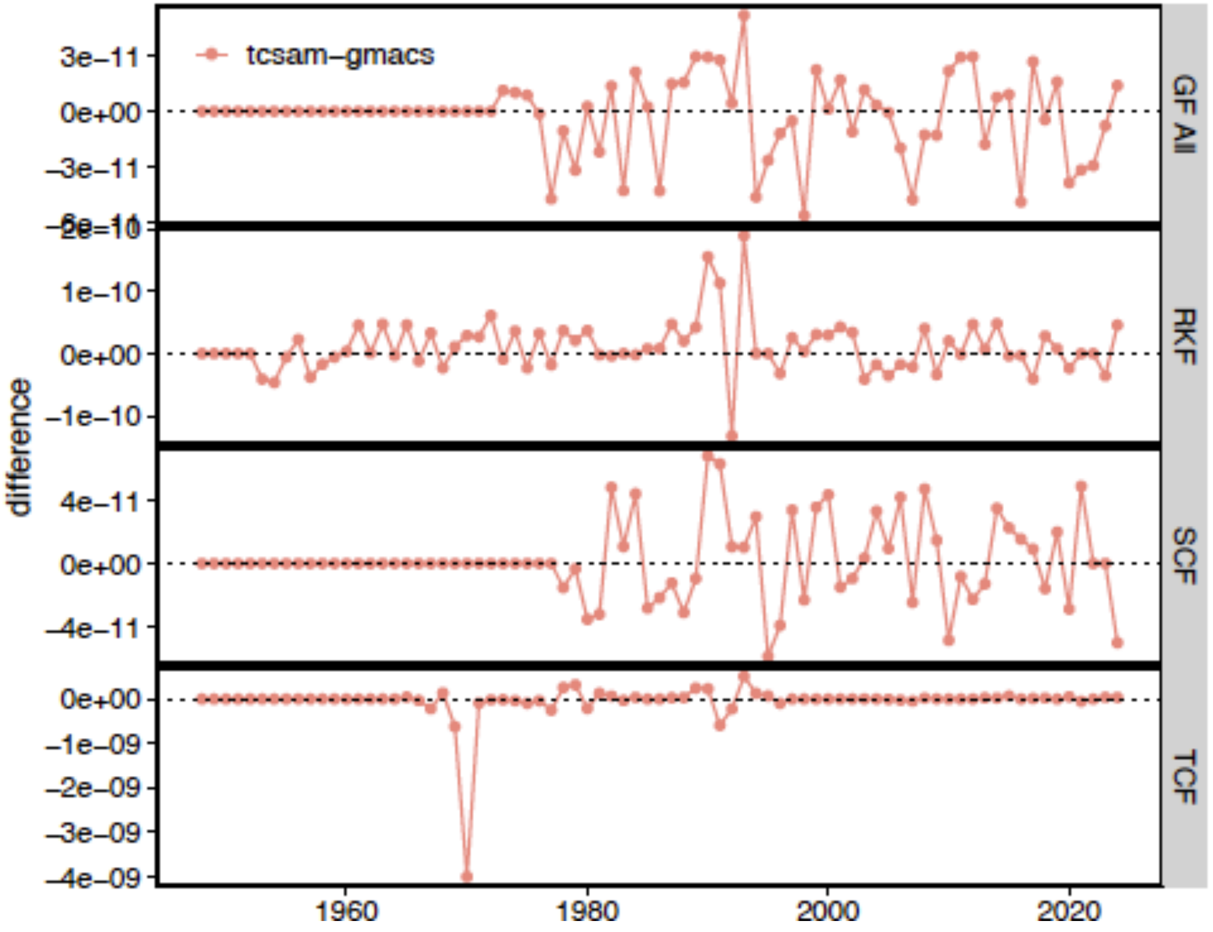
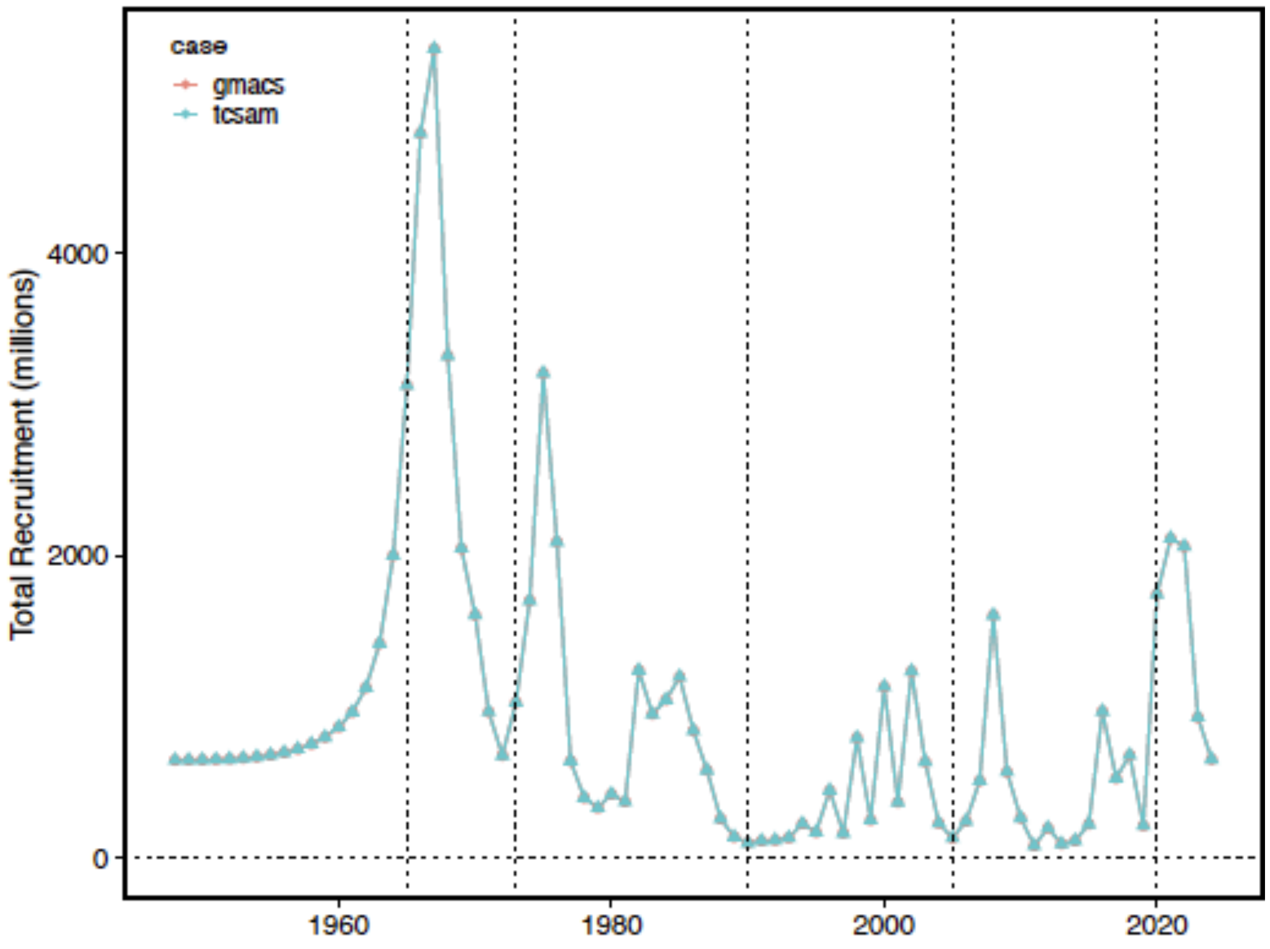
% difference
0e+00
−1e−07
−2e−07
50
100
150
post-molt size (mm CW)
%(tcsam−gmacs)

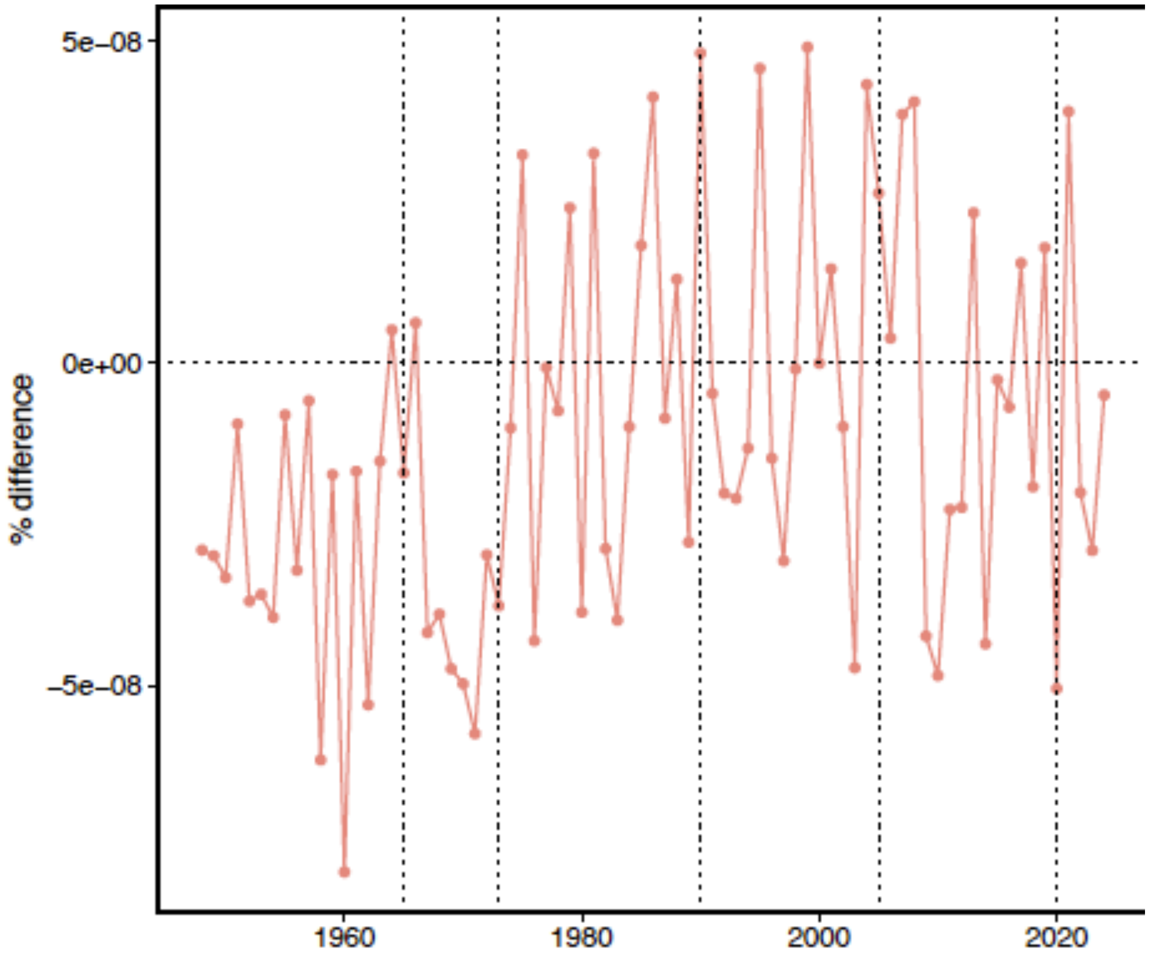
Fully-selected Male Capture F's



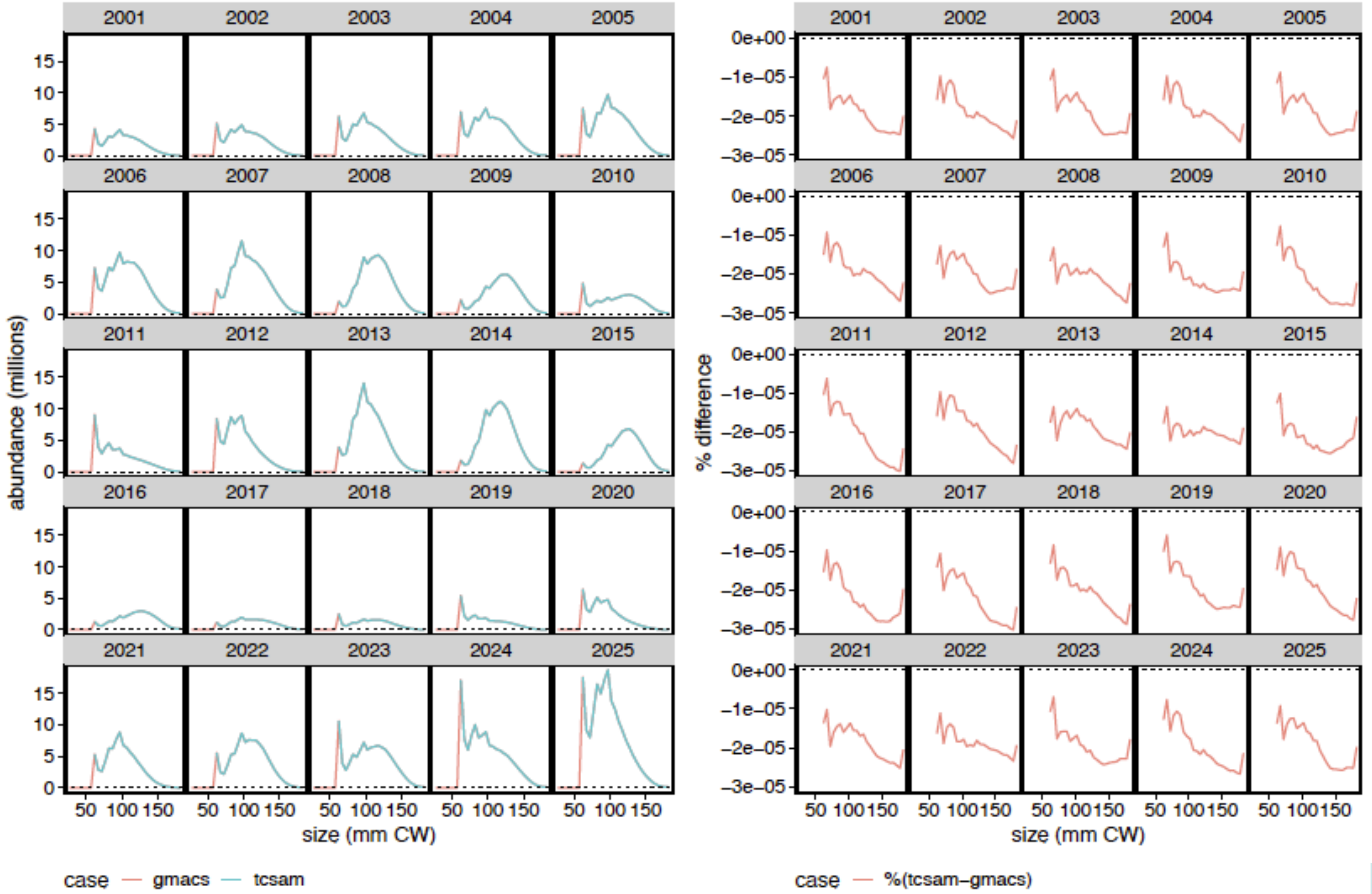


Recruitment

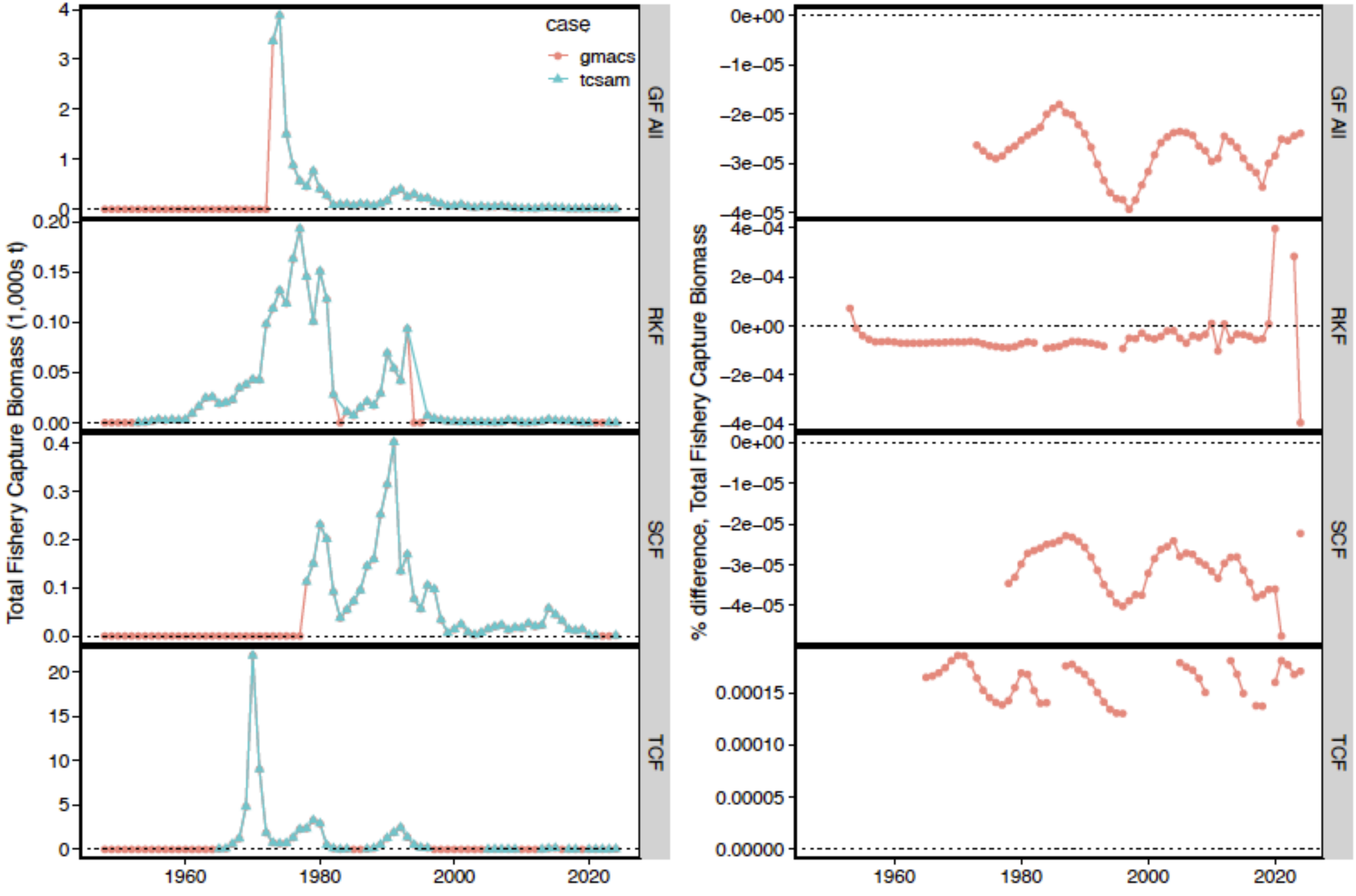




Population abundance-at-size for mature males

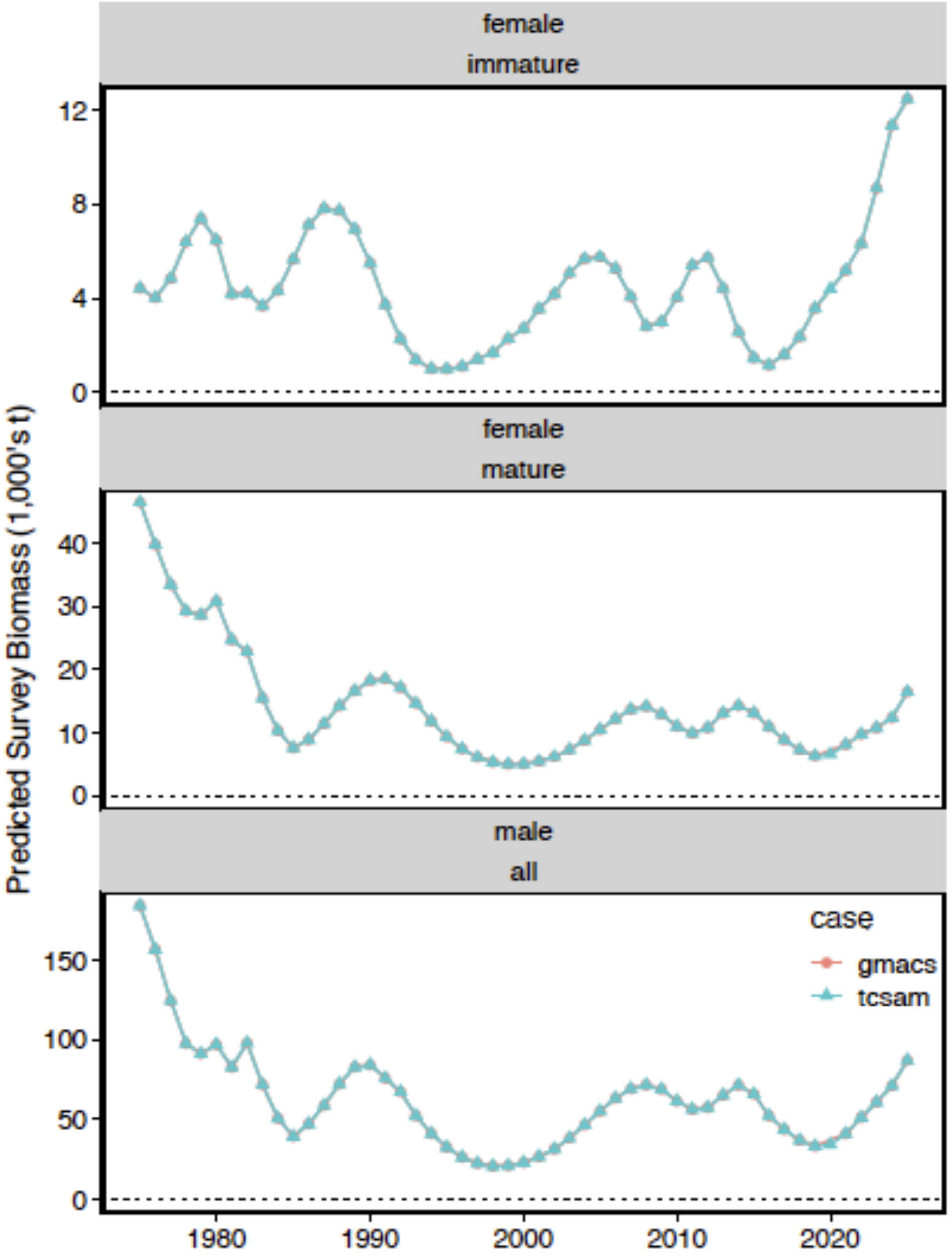


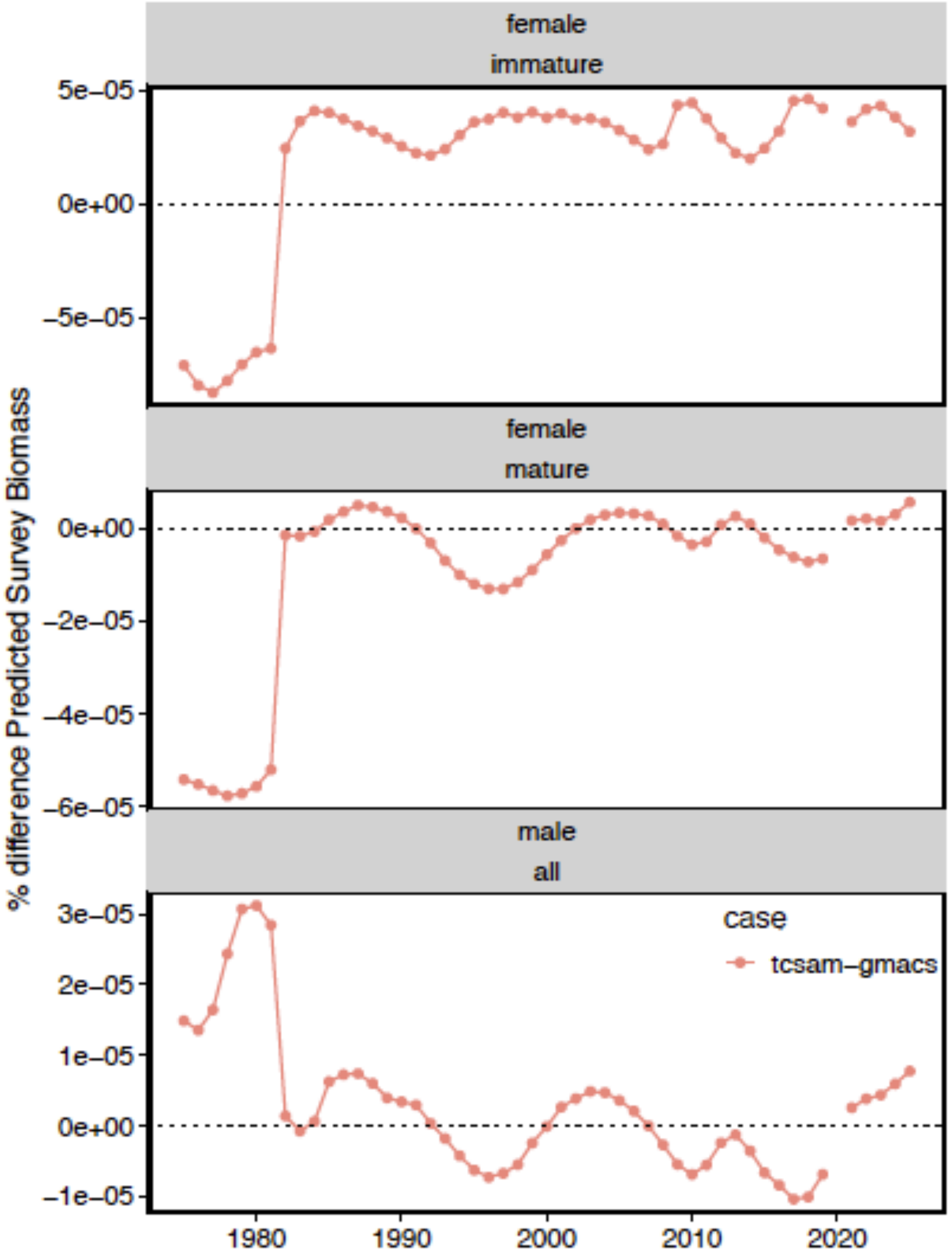
Predicted total catch biomass, by fishery



NOAA

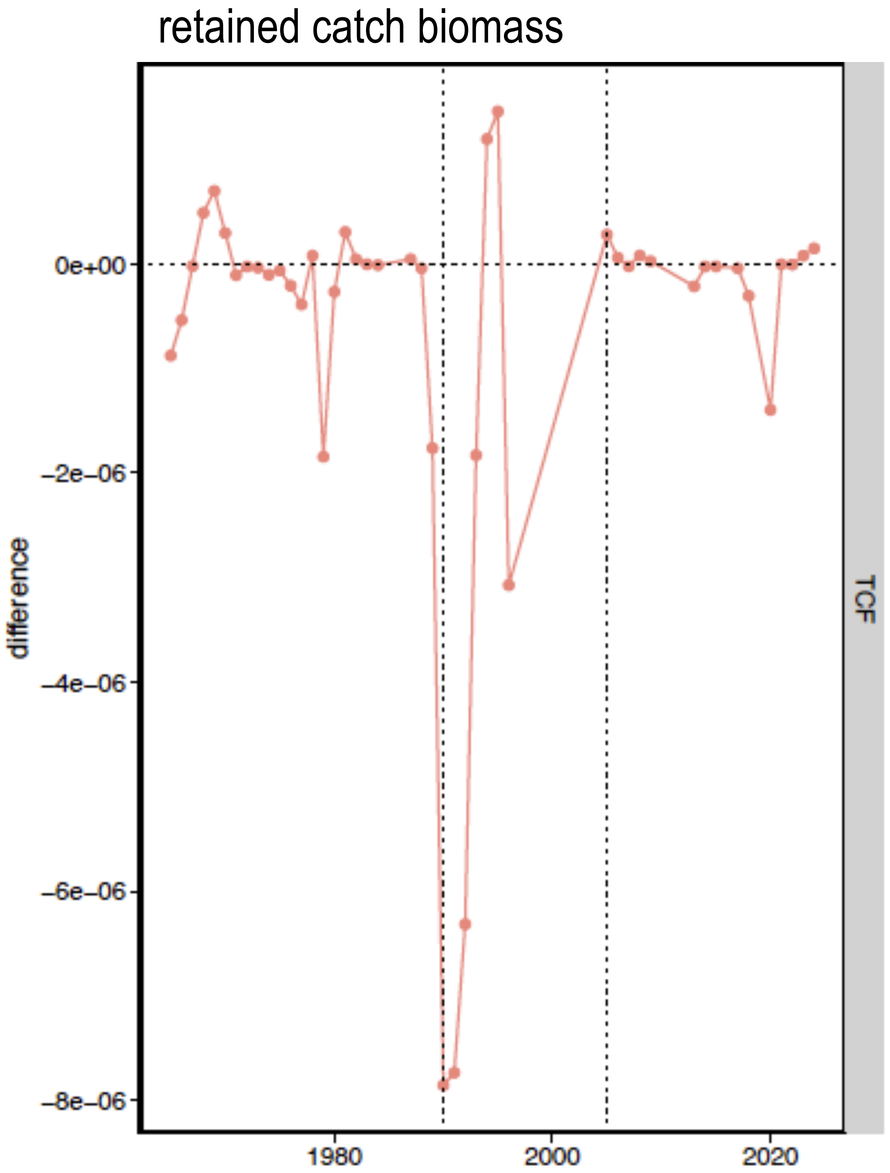
Predicted NMFS Survey Biomass

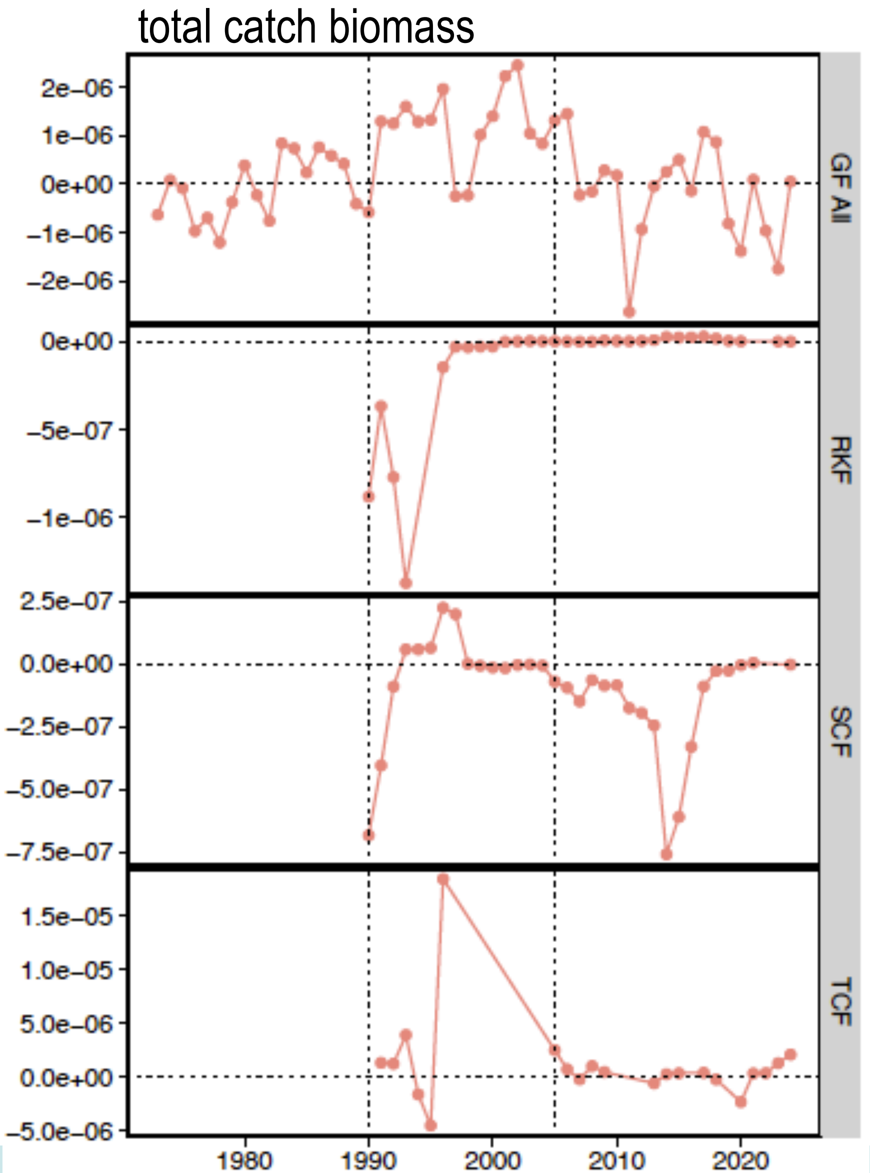




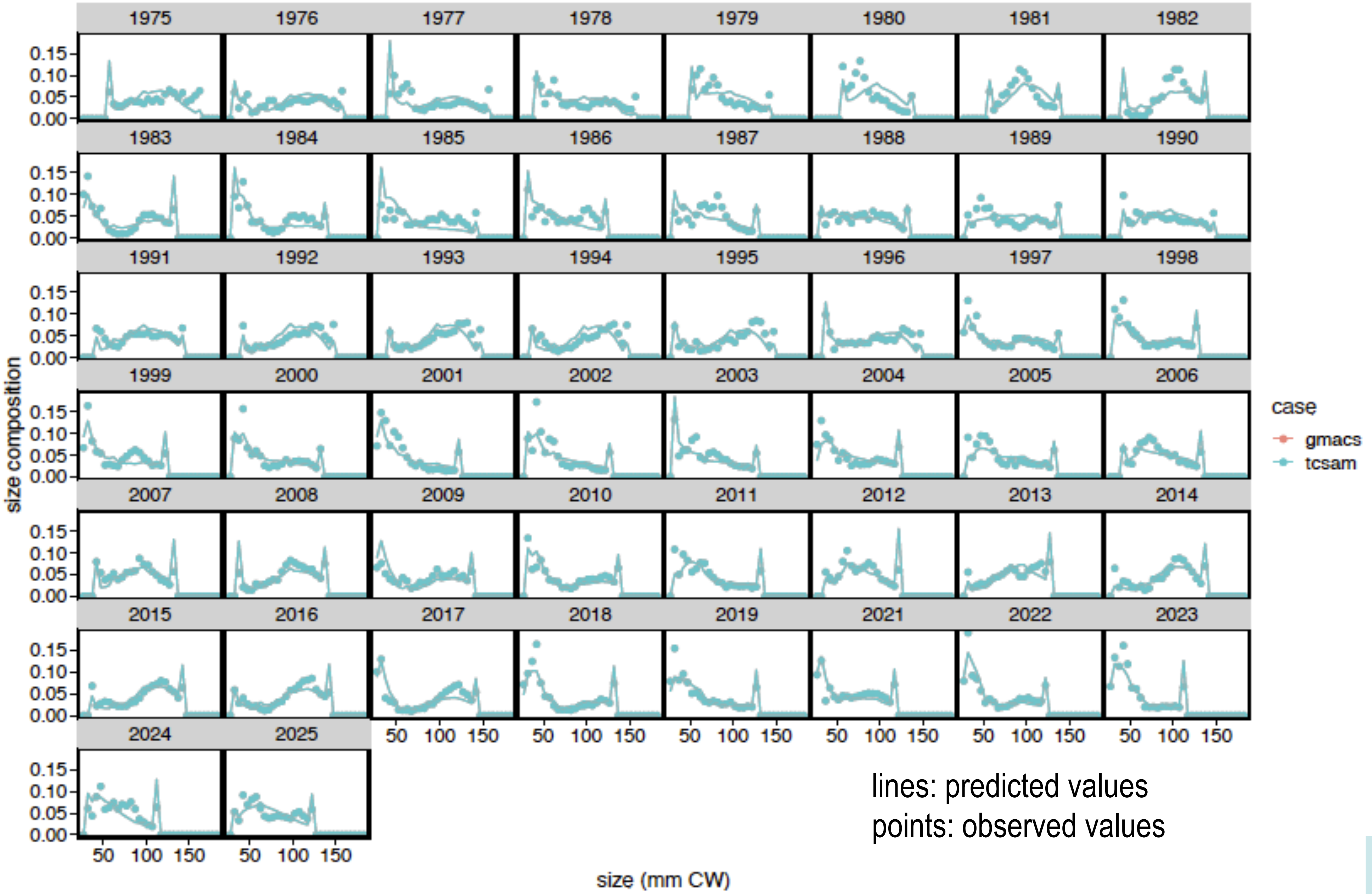
Likelihood differences

Likelihood differences



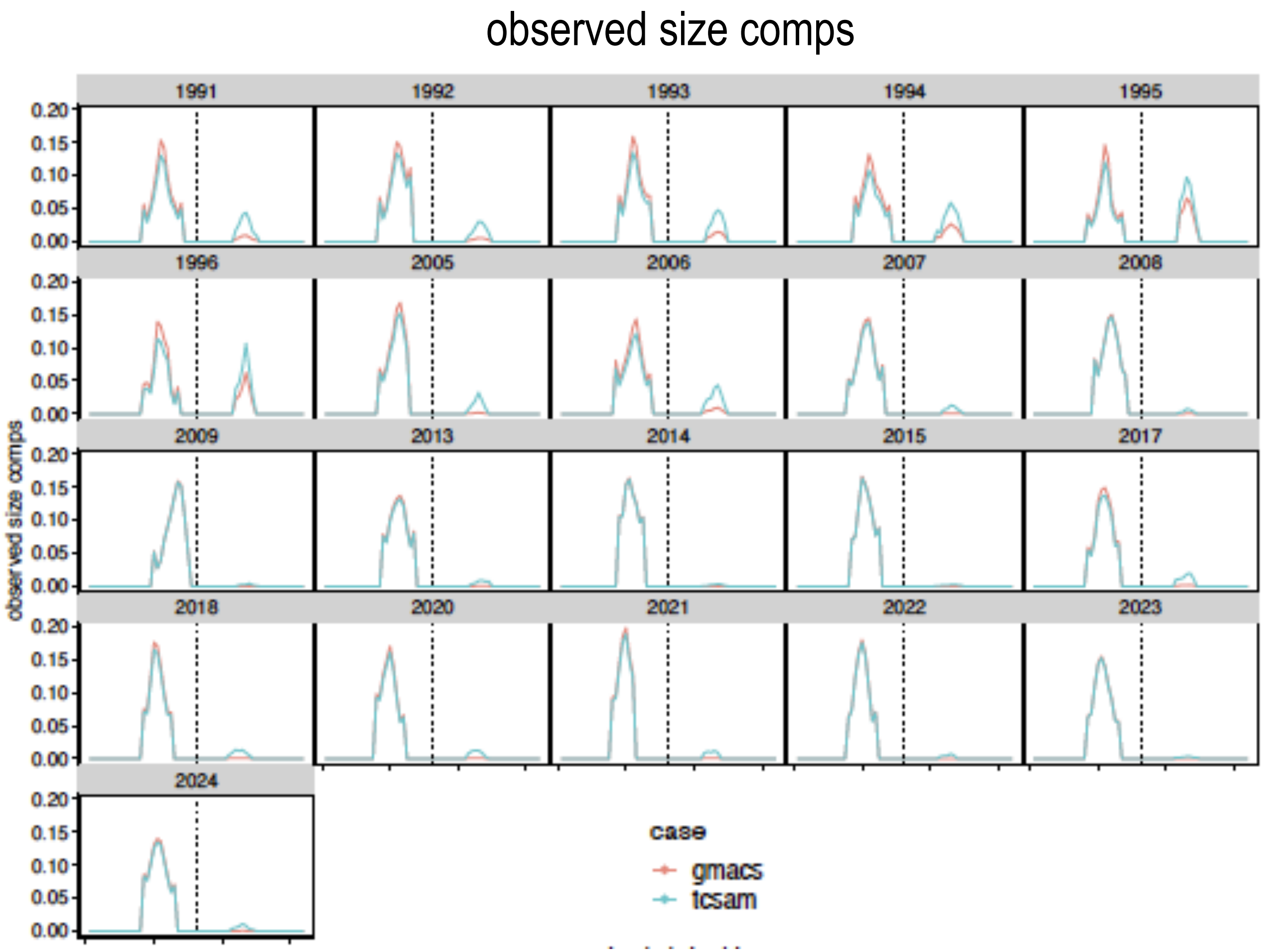


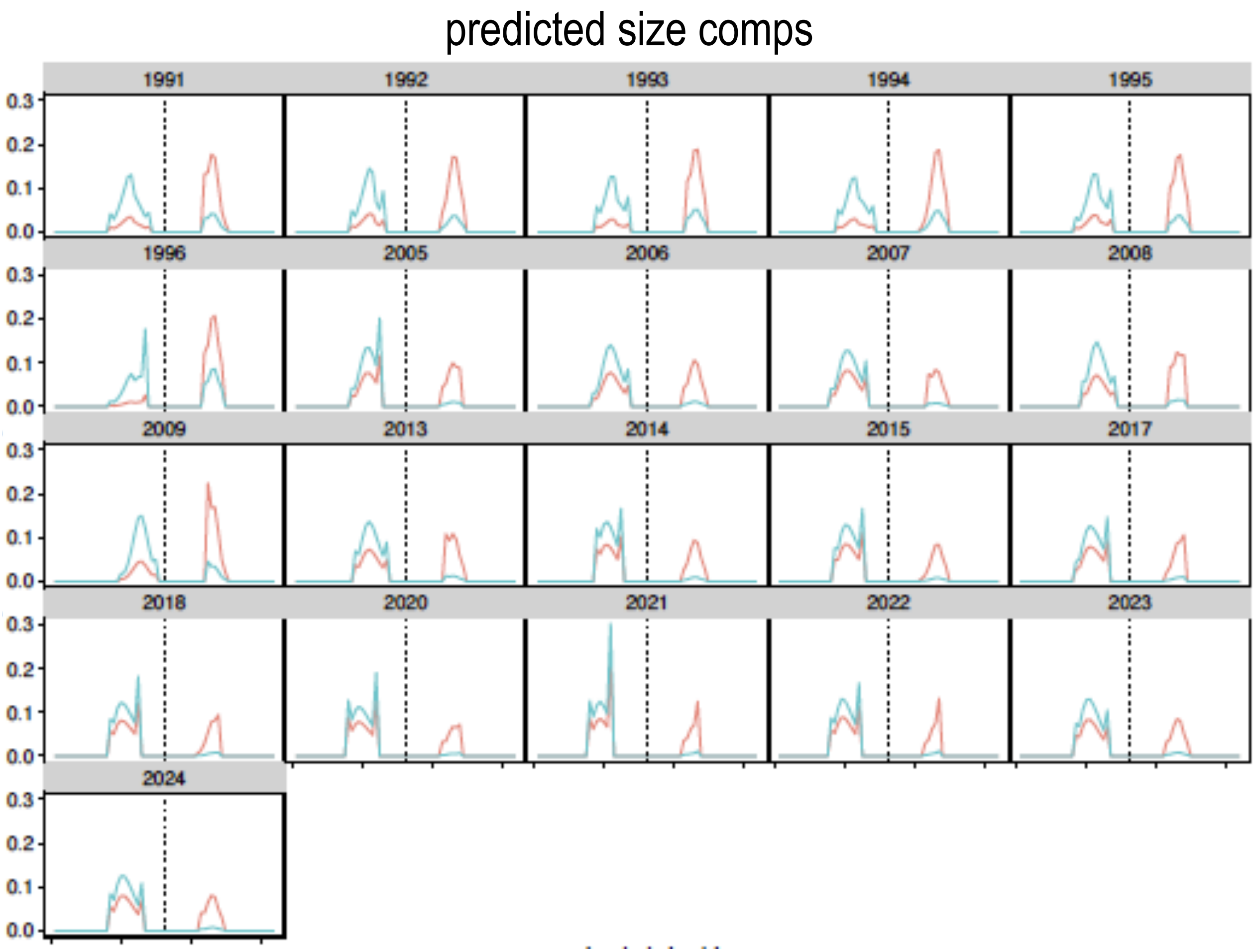
Fits to NMFS survey male size comps



lines: predicted values
points: observed values

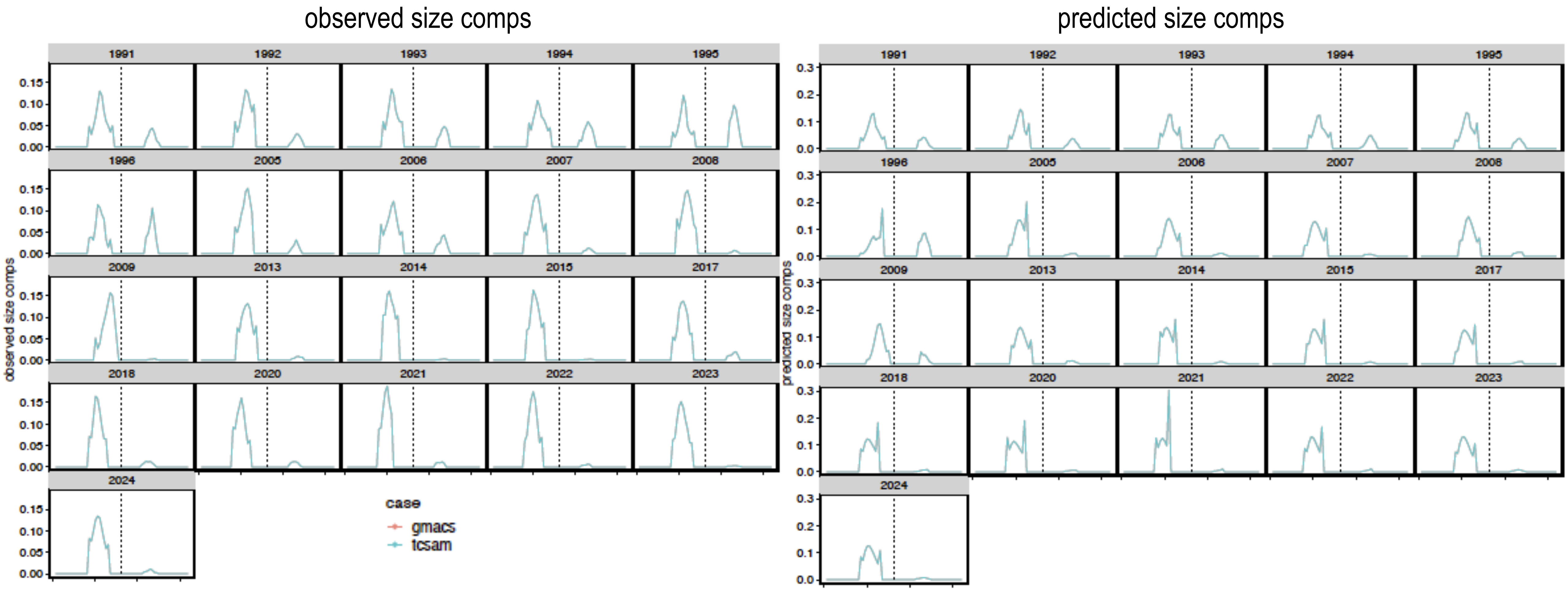
Directed fishery "extended" total catch size comps





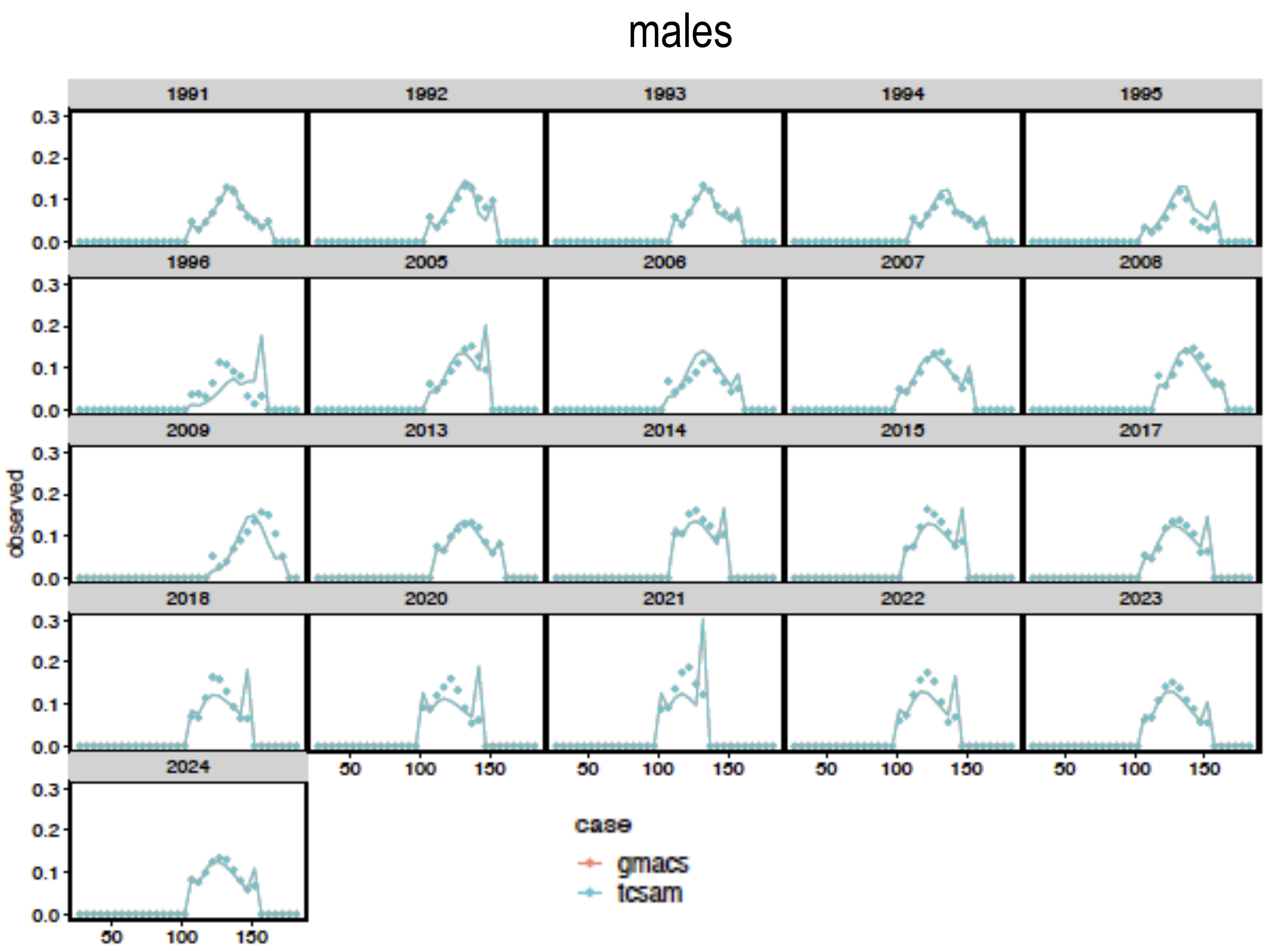
- **without scaling by sex-specific fishery capture rates**

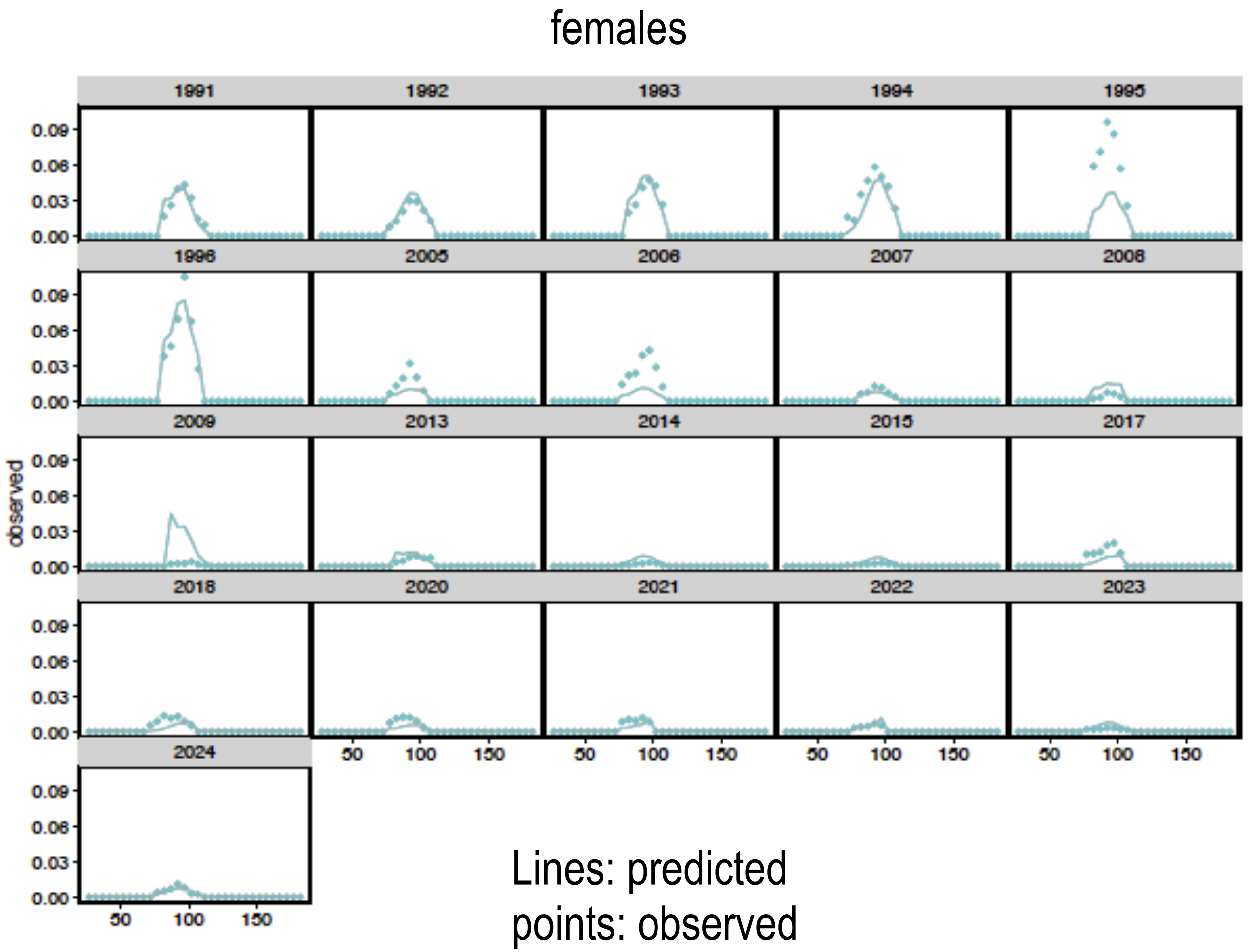
Directed fishery "extended" total catch size comps



- with scaling by sex-specific fishery capture rates

Fits to directed fishery total catch size comps

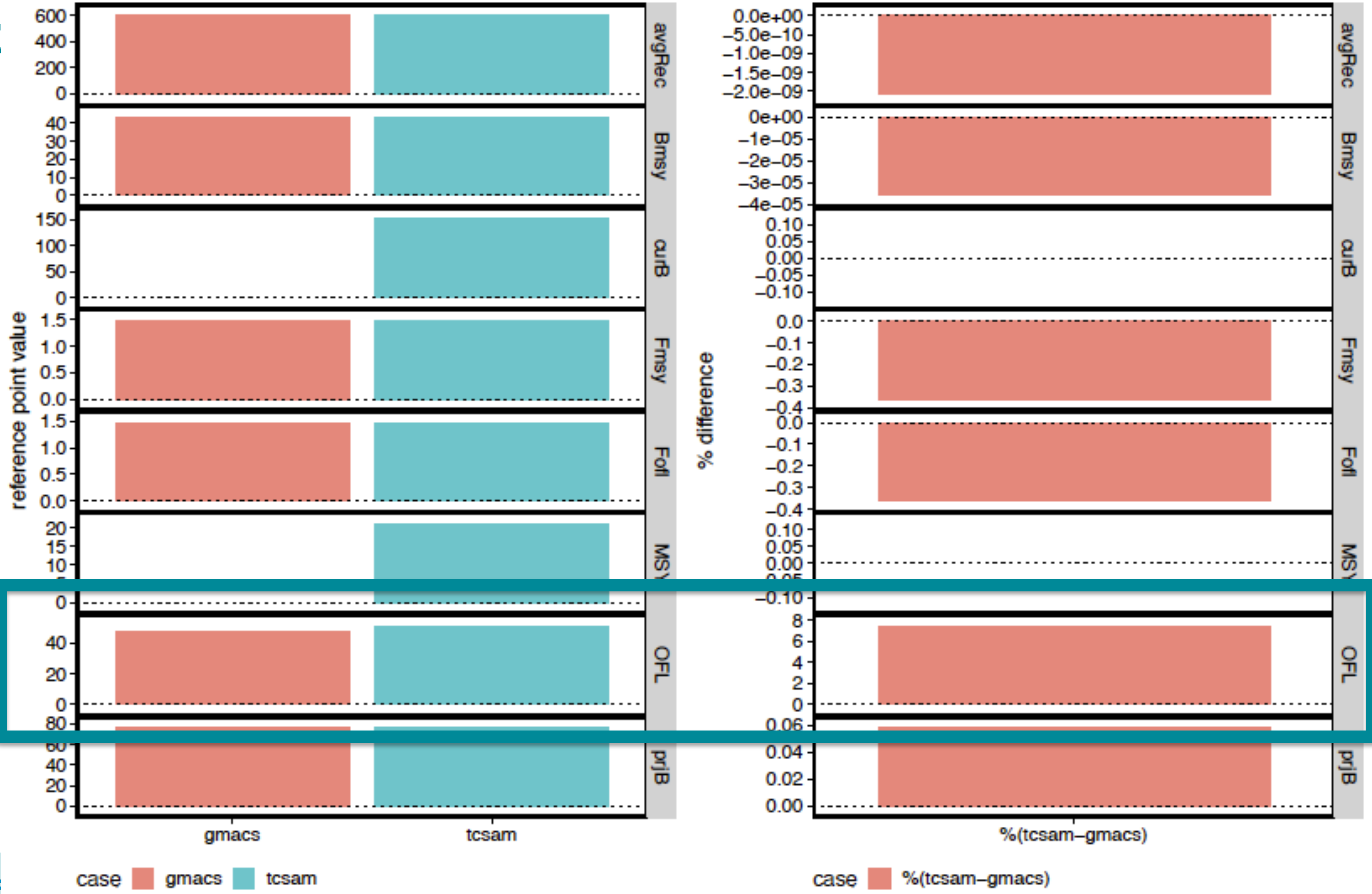




Lines: predicted
points: observed

Management Quantities

Old Results



OFL Resolution

GMACS

- uses average fleet/sex specific
  - fully-selected capture rates
  - “vulnerability”

$$\bar{v} = \overline{s \cdot (r + d \cdot (1 - r))}$$

$$F = \bar{f} \cdot \bar{v}$$

TCSAM02

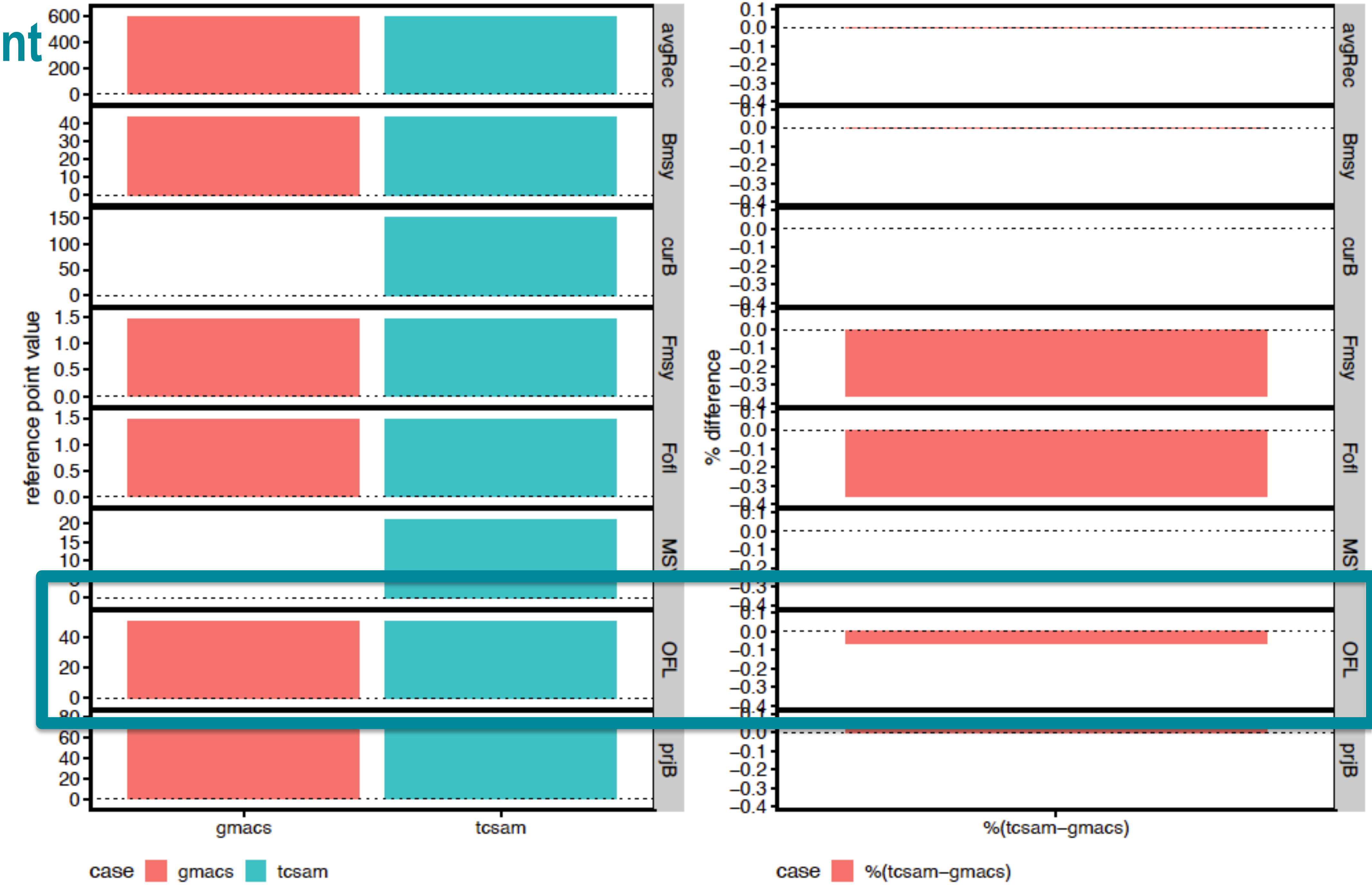
- uses average fleet/sex specific
  - fully-selected capture rates
  - selectivity
  - retention probability

$$F = \bar{f} \cdot \bar{s} \cdot (\bar{r} + d \cdot (1 - \bar{r}))$$ default

$$F = \overline{f \cdot s \cdot (r + d \cdot (1 - r))}$$ alternative

Management Quantities

New Results



Wrap-up

GMACS Estimation

- GMACS model converges, but not to TCSAM02 MLE
- Need to ensure priors and penalties are equivalent between the frameworks

Reviewers

- how close if “good enough”?
- other thoughts?